

YOURSTORY



snowflake

H2S



COCO CLI

HACKATHON

GCC Edition



Team Name: [TRidents](#)

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Team Size: 4

Problem Statement: Predictive Maintenance and OEE Command Center

PNEUMORA

EARLY AIR-LEAK PREDICTION ON REAL
COMPRESSOR SENSORS

Nine real failures. Three compressors. A breakdown predicted hours ahead.

01 · PROBLEM BRIEF

A leaking compressor takes a train out of service.

REAL

MetroPT sensors
pressure · current
oil temp · load

EXISTING ALARM

3 of 7

air leaks flagged ≥ 2 h before removal
0.49 false alerts per healthy day

LABELLED

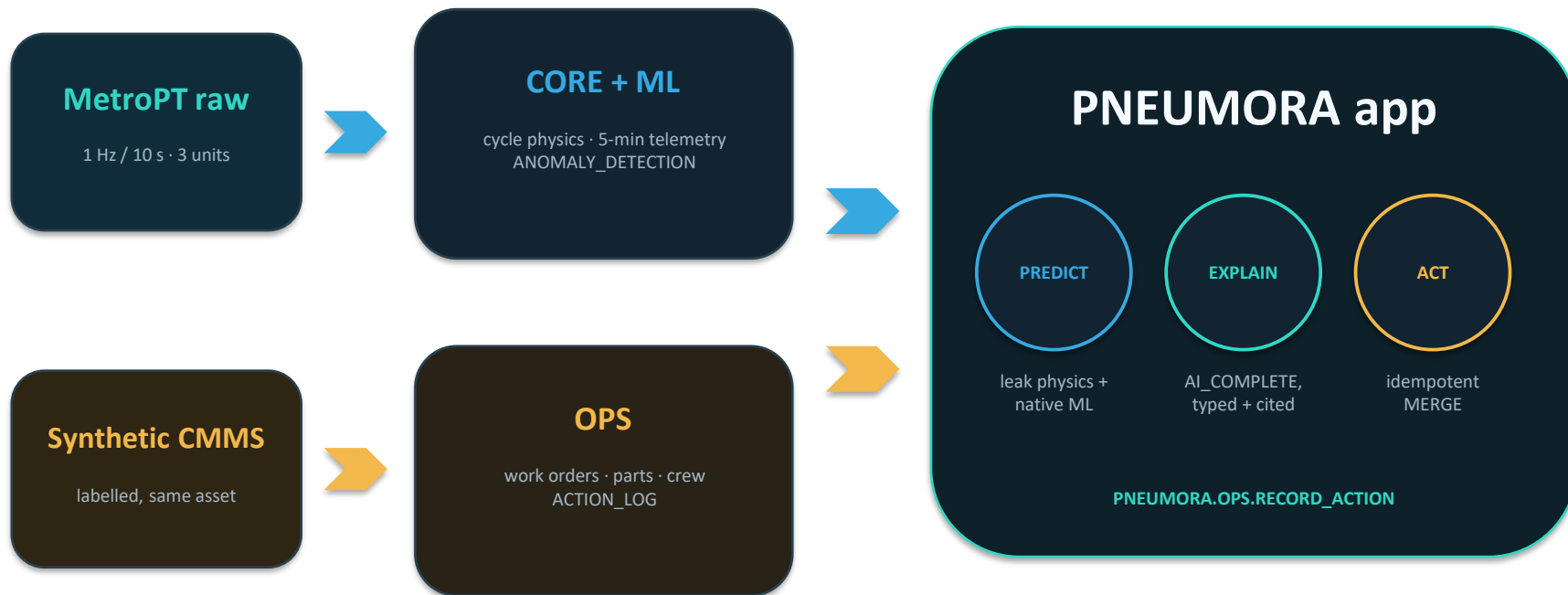
synthetic work orders
parts · crew
example-factory OEE

Same asset, one thread

PNEUMORA reads the compressor's real sensor streams, predicts 6 of 7 air-leak breakdowns hours ahead with about a ninth of the alarm's false alerts, and turns each one into a cited, recorded maintenance decision.

02 · ARCHITECTURE

Raw sensor cycles in. A recorded decision out.



03 · MEASURED RESULTS

It predicts the breakdown hours ahead. Promoted.

6 of 7**Air leaks predicted ≥ 2 h ahead**

Held out by compressor · existing alarm 3 of 7
· 0.056 vs 0.49 false alerts/day

Snowflake native ML + gradual leaks

ANOMALY_DETECTION: 4 of 4 in time.
Injected 2× air-loss leak caught 28 of 30
(control 7).

5.8 h**Median warning before removal**

PROMOTED_CROSS_VALIDATED · all five pre-
declared gates passed · chance $p = 3e-7$

What we generated

Replayed onsets and physics-injected
leaks, all **SYNTHETIC_TRAINING_ONLY**,
never used to score.

1 of 9**Warned before the leak began**

Leaks start abruptly · best of six approaches · it
predicts the breakdown, not the leak

Caveats we show on screen

Protocol v2 written after v1 failed on
false alerts; no untouched data left. No
oil leaks, RUL or savings.

The 4-minute proof: predict → explain → act



PREDICT

CoCo reads failures, alerts and native-ML results. The app shows the held-out prediction record.



EXPLAIN

Observed F04 facts go to Cortex. Every number cites its field; no invented parts.



ACT

The crew opens an inspection. A Snowflake MERGE writes one reload-safe row.

BLACK-BOX RESULT

same decision → same action id → second call deduplicated=true

YOURSTORY



1125



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THANK YOU

LIVE [PNEUMORA](#)

CODE github.com/ashokjangam/snowcore-pdm

Predict the breakdown. Explain with evidence. Record the decision. Say where prediction ends.